GHANA

- Market Functionality Index –

NOVEMBER 2024



COUNTRY MARKET ASSESSMENT : MFI IN A NUTSHELL

THE MARKET FUNCTIONALITY INDEX

The Market Functionality Index methodology is WFP's standardized market assessment approach that aims to quantify market functionality in a score per marketplace, while also identifying weaknesses or areas of interventions to guarantee the continuity or effectiveness of the assistance.

SURVEY

The methodology relies on a standard survey where questions are organized under nine dimensions deemed crucial to make a judgement statement around market functionality. These are: Assortment, Availability, Price, Resilience of Supply Chains, Competition, Infrastructure, Service, Food quality and Access and Protection.

SCORING

Markets are thus globally scored on a scale from 0 to 10 to allow worldwide comparability, where full market functionality (MFI=10) refers to a fully developed, efficient and functional market. In most contexts where WFP operates, we do not expect such scores to be achieved.

WEIGHT

The MFI considers three basic categories:

1) Cereals
2) Food items other than cereals
3) NFIs.

Cereal foods are considered more valuable by WFP to beneficiaries and thus weighted higher than non-cereal foods and NFIs. This means scores will be impacted more by the measurements (assortment, availability, etc.) of cereals than non-cereal food and NFIs.

DATA COLLECTION

Data collected 30th October to 7th November 2024

xxx traders interviewed 33 assessed marketplaces

REPORT FORMAT

Report findings have been divided into 9 key dimensions of a functioning market:

1. Assortment
2. Availability
3. Prices
4. Supply Chain Resilience
5. Competition
6. Infrastructure
7. Service
8. Food Quality
9. Access and Protection

MFI 2024 : EXECUTIVE SUMMARY

- 32 Markets Assessed in 4 regions (Brong Ahafo, Northern, Upper West and Volta).
- Data collection period: 26 Oct. – 2 Nov. 2024

KEY FINDINGS

- For the 4 regions that were assessed in this round, the **overall MFI score was 4.6** out of 10.
- The **main drivers** that brought the **overall score down** were the **Price** (3.1) and **Service** (3.8) **dimensions**. The dimensions that had the **highest average scores** were **Access and Protection** (9.8), **Competition** (9.6) and **Assortment** (9).
- **Regional average MFI scores** ranged between **4.8 (Brong Ahafo)** and **4.1 (Volta);** however it must be noted that the number of markets assessed differed by region.
- In terms of risk levels for CBT operations, 23 out of the 32 assessed markets (**72%**) were considered to be of "**high risk**" given their level of market functionality, and 5 out of 32 (**16%**) were considered to be "**medium risk**" and 4 out of 32 (**13%**) were considered to be "**low risk**."

RECOMMENDATIONS

- Given the high risk of most markets for CBT operations, various mitigation measures will need to be considered if implementing CBT operations, with one of the main ones being **close price monitoring across nearly all markets** as traders reported price increases and volatility.
- Mitigation measures for **Service** will also need to be considered for many markets, such as setting up retailer capacity strengthening training as part of market development activities.
- For markets which scored low on the **Infrastructure** dimension, possible mitigation measures include contracting traders operating in premises with adequate standards, supporting smaller retailers to obtain licenses to ensure food quality is up to standards and rehabilitating market infrastructure in the longer term.
- For markets which scored low on the **Availability** dimension, possible mitigation measures include hybrid modalities to complement missing items, setting up preferred supplier agreement or the use of vouchers to allow contracted traders to rely on secured business flow, thereby allowing the positioning of adequate stocks in advance.

COUNTRY MARKET ASSESSMENT : CONTEXT

COUNTRY CONTEXT

In 2024, Ghana continues to grapple with the lingering effects of its economic crisis and their implications for food security and poverty. During the 2024 lean season (June–August), an estimated 1.05 million people were food insecure, highlighting the vulnerability of many households. The economic crisis, rooted in the twin shocks of the COVID-19 pandemic and the Ukraine crisis, has been further exacerbated by climate-induced events, including floods and an extended dry spell. Public debt remains high, nearly equivalent to the country's annual GDP. While Ghana secured a $3 billion IMF loan in May 2023, which provided temporary relief and stabilized the Ghanaian Cedi, the recovery remains fragile.

DRY SPELL AND ATTENDANT IMPACT ON MARKETS

In 2024, inflation remains elevated, though slightly lower than the peaks of 2022-2023. Food inflation continues to pressure vulnerable households, particularly in northern Ghana, where the effects of an extended dry spell have compounded the crisis. The dry spell has significantly disrupted agricultural production, affecting over 900,000 farmers and leading to severe crop losses. The regions most affected—Northern, Upper West, and Savannah—account for a significant proportion of the country's grain production, further straining national food supplies.

The government estimates that over 900,000 farmers have been impacted by the dry spell, with 1.8 million hectares of farmland affected and total agricultural losses projected to exceed GHS 8.5 billion. The extended drought has severely curtailed agricultural output, compounding the economic difficulties faced by farming households. Ghana's social protection mechanisms, including the LEAP program, have been scaled up with WFP support to reach the most vulnerable populations, but funding shortfalls remain a critical challenge to adequately addressing food insecurity in 2024.

The availability of key food commodities, particularly maize, sorghum, millet, rice, roots, tubers, and legumes, is increasingly constrained across Ghana's markets, with shortages most acute in regions like Upper West and Oti. Nearly three-quarters of traders expect maize stocks to be depleted within a week, and many are unable to accommodate increased demand, highlighting the fragility of market supplies. Price volatility is escalating, with traders reporting significant increases in the prices of cereals (e.g., maize up 97%) and non-cereal staples due to supply-demand imbalances exacerbated by a prolonged dry spell. This dynamic has reduced purchasing power and consumer demand, further destabilizing markets and deepening food insecurity. Market projections indicate continued price hikes through late 2024 due to delayed harvest impacts, threatening the stability of supply chains and pushing vulnerable populations into greater hardship.

COUNTRY MARKET ASSESSMENT : METHODOLOGY

This survey is an in-depth assessment, where 17 trained enumerators will conduct face-to-face interviews and key informants interviews. The survey is in two folds, the trader survey and the market survey. For the market survey, 18 traders will be purposively sampled in each market for interview through the following heterogeneous cases:

- Market chains: cereals, non-cereal food, and non-food items (NFI)
- Trader function: retail and wholesale/collector. Collectors buy directly from farmers and sell to wholesalers.
- Trader volume size: small, medium and large.

For market survey, 18 key informants will be interviewed in each of the 32 markets of interests.

Data Collection

17 trained enumerators will be deployed to the field to collect data in all 34 markets over a period of 4 days using the standard Market Functionality Index Full tool (v1_0). This standard tool contains both the market survey and trader survey forms.

--> https://enketo.moda.wfp.org/x/RpPRvIlA

Over the 4-day period of data collection, a total of 614 traders and 34 key informants will be interviewed. Two enumerators will visit one market per day and each of them will interview nine (9) traders in a market (2x9=18 traders in each market), which implies that a total 36 interviews will be conducted in two markers daily. Enumerators will be supervised by WFP staff in the first three days of data collection.

DIMENSION SCORE

MFI rate between zero (low functionality) and ten (high functionality). The MFI is a combination of these intermediate scores, ranging from zero to ten as well. The common scale allows to compare a wide range of marketplaces, some of which barely, others highly functional. The analysis of intermediate scores remains insightful to disentangle the drivers of a market's functionality and identify possible risks and opportunities associated with market-based interventions.

Region	District
Northern	Tolon
Northern	Zabzugu
Northern	Mion
Northern	Yendi
Oti	Krachi East Municipal
Oti	Nkwanta South Municipal
Savannah	Bole
Savannah	Central Gonja
Savannah	East Gonja Municipal
Savannah	North Gonja
Savannah	Sawla-Tuna-Kalba
Savannah	West Gonja
Upper West	Jirapa
Upper West	Wa Municipal
Upper West	Wa West
Upper West	Daffiama Bussie Issa
Upper West	Wa East

Region	District
Bono	Banda
Bono	Jaman North
Bono	Jaman South Municipal
Bono	Tain
Bono East	Kintampo North Municipal
Bono East	Kintampo South
Bono East	Nkoranza North
Bono East	Pru
North East	Mamprugu Moagduri
Northern	Kumbungu
Northern	Nanumba North
Northern	Nanumba South
Northern	Savelugu
Northern	Tamale Metropolitan
Northern	Tatale Sanguli

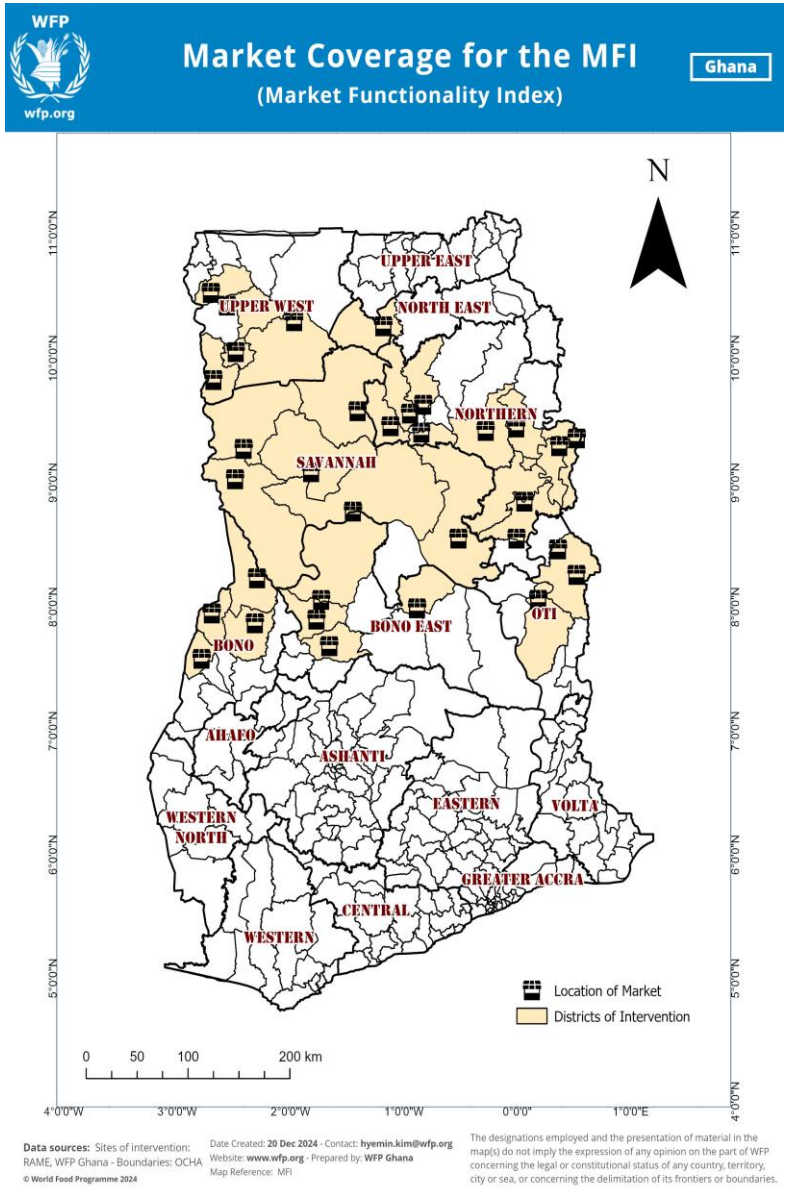
COUNTRY MARKET ASSESSMENT

MOTIVATION OF THE STUDY

Several regions in Ghana experienced an extended drought, which has drastically reduced agricultural output, intensifying economic hardships for farming households and straining Ghana's food supply. Market availability of key staples—maize, sorghum, millet, rice, roots, tubers, and legumes—is tightening, especially in Upper West and Oti, where nearly 75% of traders expect maize stocks to run out within a week. Rising demand and supply shortages have triggered significant price surges, with maize prices up 97%, weakening purchasing power and exacerbating food insecurity. Market projections indicate continued price hikes through late 2024, further destabilizing supply chains and deepening vulnerabilities.To forestall further deterioration of the food security in these the most affected areas, WFP will provide unconditional cash assistance to alleviate the brunt of economic hardship and food insecurity, especially amongst households headed by, or with vulnerable members such as pregnant and breastfeeding women and people living with disabilities. This assessment was needed to ascertain the choice of transfer modality and provide recommendations for CBT operations.

DEFINITIONS

The **assortment** of essential goods dimension indicates which classes of goods to meet essential needs can be purchased in a marketplace and how much choice is offered. The **availability** dimension assesses whether certain products are scarce or likely to get scarcer in the short run. The **price** dimension assesses price trends and volatility. Both rising prices and excessive volatility show that the market is not functioning well. The **supply chain resilience** dimension evaluates both responsiveness and vulnerability of supply chains. Resilient supply chains underpin the regular supply of a market with goods despite potential disruptions, which is essential to a well-functioning market. The **competition dimension** evaluates if the number of traders in the market and the distribution of power among them guarantee a reasonable level of competition. Competition is critical to a well-functioning market since it forces vendors to improve their efficiency and pushes prices down to the benefit of the end consumer. The **infrastructure** dimension assesses the type and condition of the physical structures which host shops in addition to, for example, sewage system, electricity, or communication network. Adequate infrastructure contributes both to a well-functioning market and can be interpreted as an indication for different aspects of market efficiency. The service dimension looks at the service provided while shopping as well as during check-out. The level of **service** is directly or indirectly associated with different features of a well-functioning market such as transparency, competition, or reliability. **Access & Protection** issues reduce to a large degree market functioning for certain groups of people (both customers and traders) or for everyone. These functionality lapses can be short-lived or prolonged. In the latter case, normally markets adapt to the new circumstances and manage to operate, even though with large degrees of inefficiency.



COUNTRY MARKET ASSESSMENT : OVERVIEW

OVERVIEW

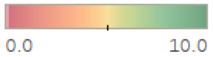
At the national level (i.e. for the 4 regions that were assessed in this round), the overall MFI score was 4.6, indicating medium-low level market functionality. The main drivers that brought the score down were price volatility and unpredictability and poor service quality*. The Brong Ahafo region had a slightly higher regional MFI score of 4.8 compared to the other regions.

In general, most assessed markets were well supplied with a good assortment of food and non-food items, and with the exception of 1 market, there was little concern for the competition dimension (Yendi market in the Northern region). With the exception of Kumbungu market in the Northern region, all markets scored a 10 on the access and protection dimensions. Overall, supply chains were also deemed resilient with relatively good food quality in markets. Scores varied on the availability and infrastructure dimensions across markets.

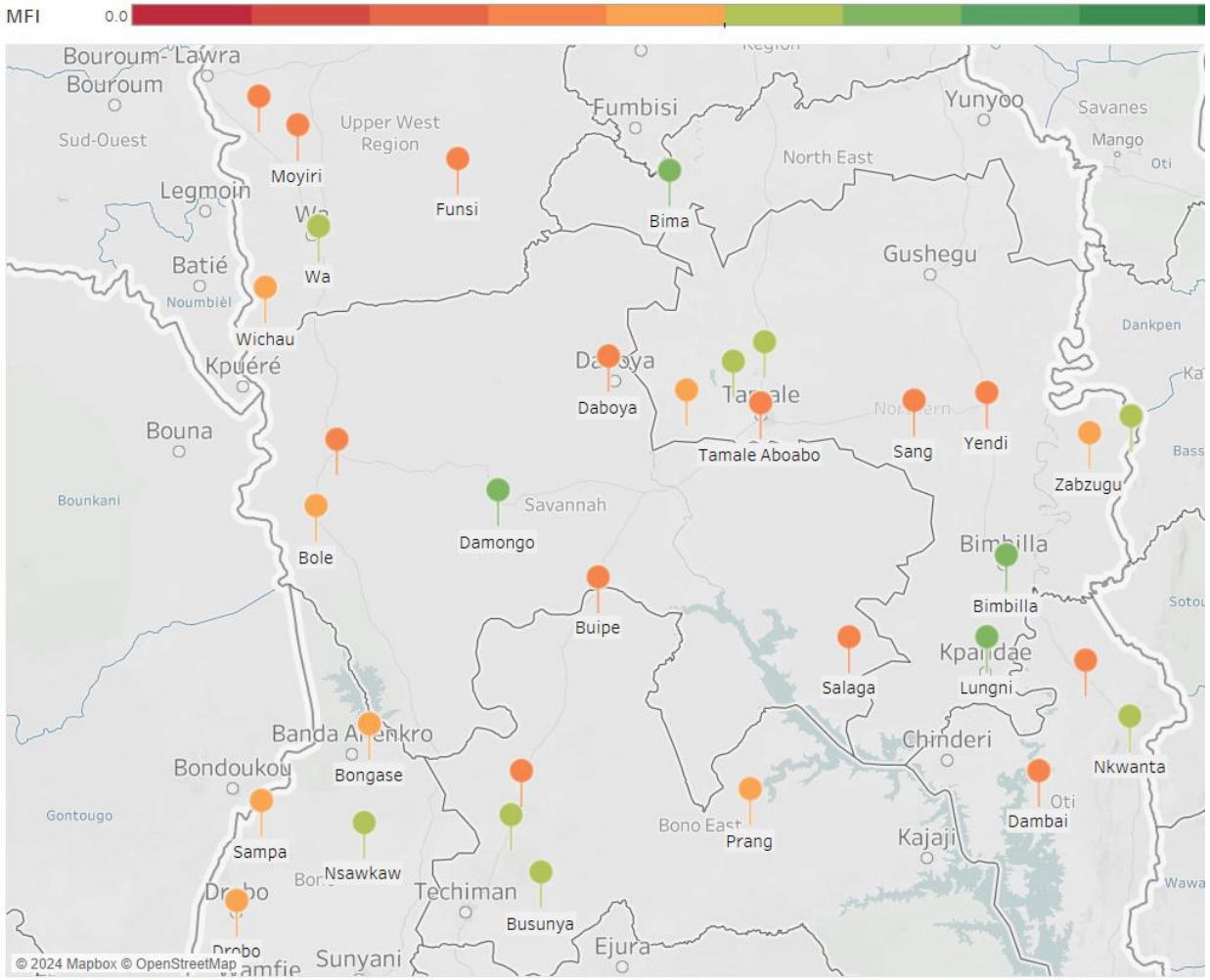
**Markets in the Western Africa region tend to score relatively low on the service dimension due to low maturity.*

Across Market comparison

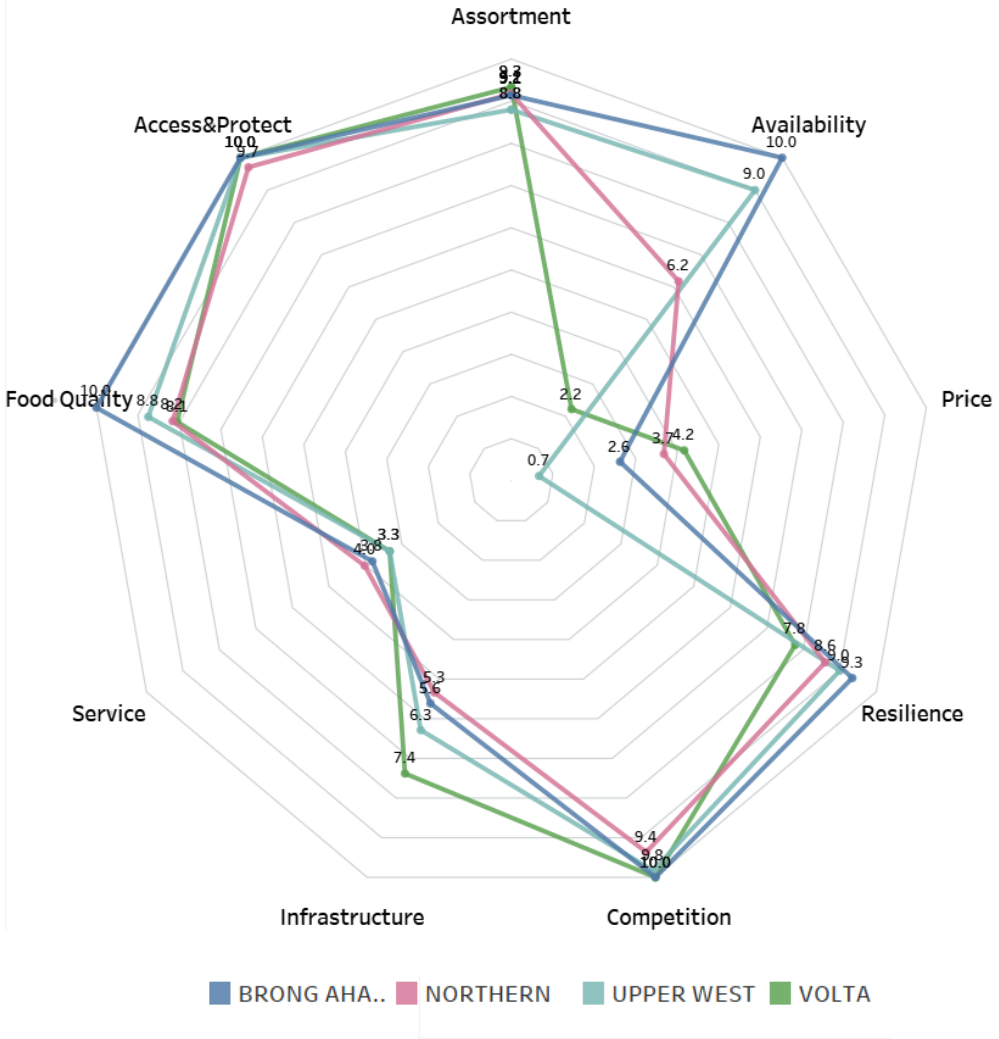
Adm1Name	MarketName	Survey ID (M-Y)	Assortm..	Availabil..	Price	Resilience	Competi..	Infrastru..	Service	Quality	AccessPr..	MFI
BRONG AHAFO	Bongase	4824, (November-2024)	6.0	10.0	5.0	10.0	10.0	2.9	1.7	10.0	10.0	4.7
	Busunya	4824, (November-2024)	10.0	10.0	2.5	9.6	10.0	3.6	5.0	10.0	10.0	5.4
	Drobo	4824, (November-2024)	9.0	10.0	0.0	10.0	10.0	8.6	5.0	10.0	10.0	4.3
	Jema	4824, (November-2024)	10.0	10.0	5.0	8.3	10.0	6.4	3.3	10.0	10.0	6.0
	Nsawkaw	4824, (November-2024)	9.0	10.0	5.0	10.0	10.0	5.0	3.3	10.0	10.0	5.9
	Prang	4824, (November-2024)	10.0	10.0	0.8	7.9	10.0	5.7	3.3	10.0	10.0	4.4
	Sampa	4824, (November-2024)	10.0	10.0	0.0	9.6	10.0	7.1	5.0	10.0	10.0	4.2
	Regional Average		9.1	10	2.6	9.3	10.0	5.6	3.8	10.0	10.0	4.8
NORTHERN	Bima	4824, (November-2024)	6.0	7.5	7.5	7.5	10.0	8.6	5.0	10.0	10.0	6.7
	Bimbilla	4824, (November-2024)	9.0	10.0	10.0	7.1	9.2	7.1	3.3	8.8	10.0	6.0
	Bole	4824, (November-2024)	10.0	0.8	5.0	7.1	10.0	7.9	3.3	10.0	10.0	4.2
	Buipe	4824, (November-2024)	9.0	10.0	0.0	9.6	10.0	4.3	3.3	5.0	10.0	3.6
	Daboya	4824, (November-2024)	9.0	10.0	0.0	8.3	10.0	1.4	3.3	5.0	10.0	3.4
	Damongo	4824, (November-2024)	10.0	10.0	4.2	10.0	10.0	8.6	3.3	10.0	10.0	6.1
	Katinga	4824, (November-2024)	10.0	3.3	1.7	8.8	10.0	2.1	5.0	8.8	10.0	4.4
	Kumbungu	4824, (November-2024)	8.0	3.3	5.0	6.7	10.0	7.1	5.0	8.8	5.0	5.2
	Lungni	4824, (November-2024)	9.0	10.0	10.0	7.5	10.0	7.1	3.3	10.0	10.0	6.2
	Salaga	4824, (November-2024)	10.0	3.3	0.0	7.5	10.0	5.7	5.0	6.3	10.0	3.5
	Sang	4824, (November-2024)	10.0	0.8	5.0	8.8	10.0	2.9	3.3	6.3	10.0	3.8
	Savelugu	4824, (November-2024)	9.0	10.0	2.5	10.0	10.0	2.9	3.3	7.5	10.0	5.1
	Sawla	4824, (November-2024)	9.0	8.3	0.0	10.0	10.0	7.1	5.0	7.5	10.0	4.0
	Tamale Aboabo	4824, (November-2024)	10.0	1.7	0.0	7.5	10.0	8.6	3.3	10.0	10.0	3.6
	Tatale	4824, (November-2024)	8.0	5.0	5.0	10.0	10.0	3.6	3.3	10.0	10.0	5.5
	Yendi	4824, (November-2024)	10.0	5.8	2.5	10.0	0.0	3.6	5.0	7.5	10.0	3.3
	Zabzugu	4824, (November-2024)	10.0	5.0	4.2	10.0	10.0	2.1	5.0	7.5	10.0	4.9
	Regional Average		9.2	6.2	3.7	8.6	9.4	5.3	4.0	8.2	9.7	4.7
UPPER WEST	Funsi	4824, (November-2024)	7.0	10.0	0.0	7.5	10.0	4.3	5.0	10.0	10.0	3.8
	Jirapa	4824, (November-2024)	10.0	8.3	0.0	10.0	10.0	7.1	1.7	7.5	10.0	3.8
	Moyiri	4824, (November-2024)	7.0	10.0	0.0	7.5	9.2	5.0	5.0	8.8	10.0	3.7
	Wa	4824, (November-2024)	10.0	6.7	2.5	10.0	10.0	7.9	3.3	8.8	10.0	5.3
	Wichau	4824, (November-2024)	10.0	10.0	0.8	10.0	10.0	7.1	1.7	8.8	10.0	4.5
	Regional Average		8.8	9.0	0.7	9.0	9.8	6.3	3.3	8.8	10.0	4.2
VOLTA	Dambai	4824, (November-2024)	9.0	0.0	5.0	7.9	10.0	8.6	3.3	8.8	10.0	3.7
	Kpassa	4824, (November-2024)	10.0	0.0	5.0	6.3	10.0	5.0	3.3	6.7	10.0	3.4
	Nkwanta	4824, (November-2024)	9.0	6.7	2.5	9.2	10.0	8.6	3.3	8.8	10.0	5.3
	Regional Average		9.3	2.2	4.2	7.8	10.0	7.4	3.3	8.1	10.0	4.1
Average			9.0	6.9	3.1	8.8	9.6	5.7	3.8	8.6	9.8	4.6



COUNTRY MARKET ASSESSMENT : OVERVIEW



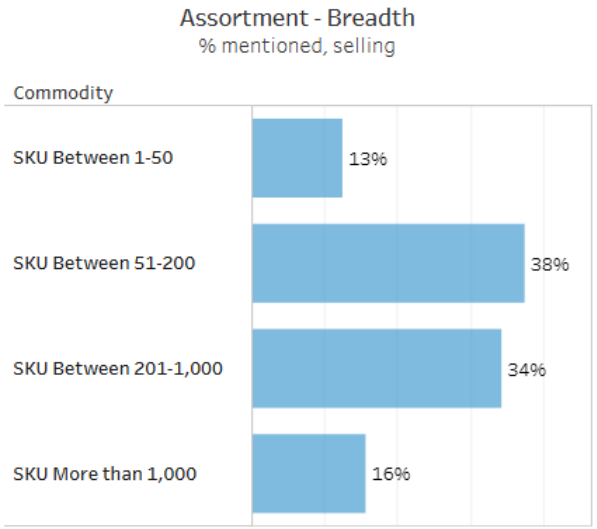
Regional Results by Dimension

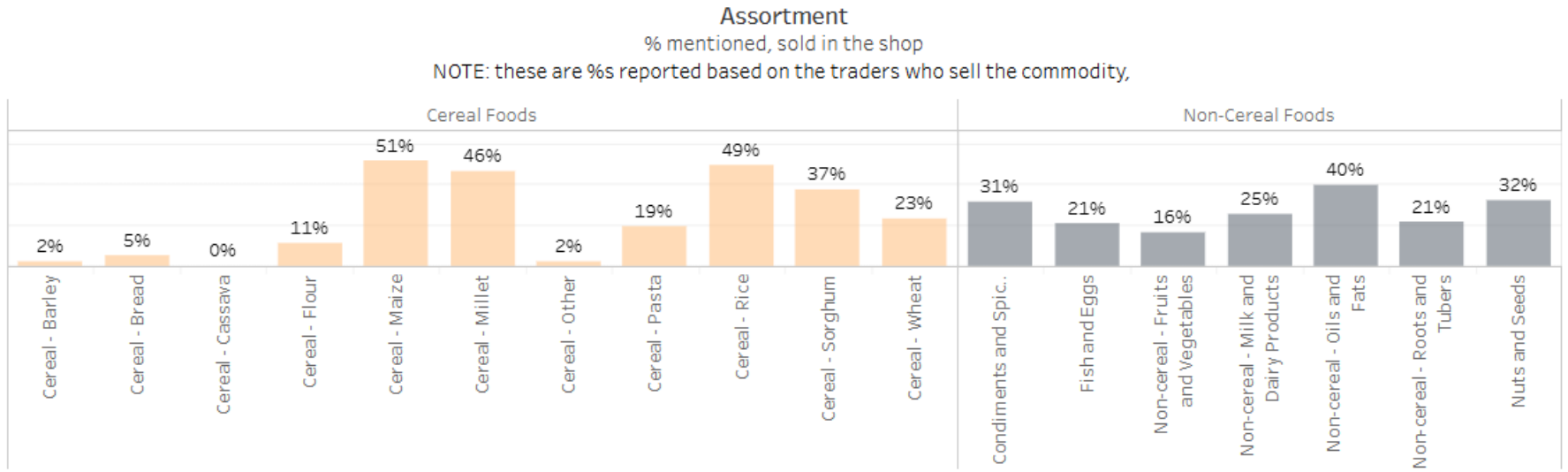


FINDINGS BY MFI DIMENSION : ASSORTMENT

There is a relatively good diversity of basic food products in the markets assessed, both in terms of cereals and non-cereal foods. In terms of NFIs, most markets carry all essential goods, however lack of health and shelter items were reported for Bongase market in Brong Ahafo region and Tatale market in the Northern region. Bima market in Northern region was flagged for having the least assortment of NFI items (communication, education, health and WASH).

The 2 markets that had the lowest score on this dimension were Bongase in Brong Ahafo and Bima market in the Northern region; they both scored 6 out of 10. Bongase's score was impacted primarily by low breadth of assortment (100% of traders assessed having SKU between 1-50) and lack of health and shelter NFIs. Bima's score was mainly brought down due to its lack of NFI assortment as aforementioned.





	BRONG AHAFO							NORTHERN																	UPPER WEST					VOLTA		
Commodity	Bongase	Busunya	Drobo	Jema	Nsawkaw	Prang	Sampa	Bima	Bimbilla	Bole	Buipe	Daboya	Damongo	Katinga	Kumbun..	Lungni	Salaga	Sang	Savelugu	Sawla	Tamale Aboabo	Tatale	Yendi	Zabzugu	Funsi	Jirapa	Moyiri	Wa	Wichau	Dambai	Kpassa	Nkwanta
Cereal Food	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Other Food	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
NFI - Communication	100%	100%	100%	100%	100%	100%	100%	0%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
NFI - Education	100%	100%	100%	100%	100%	100%	100%	0%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
NFI - Household Items	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
NFI - Health	0%	100%	100%	100%	100%	100%	100%	0%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	0%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
NFI - Shelter	0%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	0%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
NFI - WASH	100%	100%	100%	100%	100%	100%	100%	0%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

FINDINGS BY MFI DIMENSION : AVAILABILITY

Availability of essential goods does not appear to be an issue in the majority of the assessed markets; all markets scored a 10 in the Brong Ahafo region, and the Upper West also had a high regional average score of 9.1. However, a few markets scored very low on this dimension in the Northern region as well as Volta region.

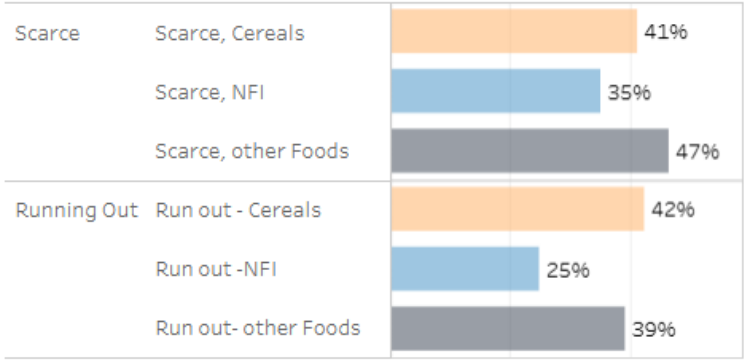
Bole and Sang markets in the Northern region both scored very low (0.8); traders in Bole reported scarcity and running out of all commodity types, especially oils and fats. In Sang, scarcity of NFIs was noted as well as rice. Dambai and Kpassa markets in Volta region scored 0, with traders reporting scarcity and running out of all commodity types (millet, sorghum and maize in particular among cereals for Dambai market and maize in particular among cereals for Kpassa market).

Tamale Aboabo market in Northern region also had a low score of 1.7, with the majority of traders reporting scarcity across all commodities. Several traders reported issues of scarcity and running out of oils and fats and rice.

Katinga, Kumbungu and Salaga markets in Northern region also scored a low 3.3, with low availability of cereals such as maize, millet and rice.

RESULTS FOR ALL MARKETS

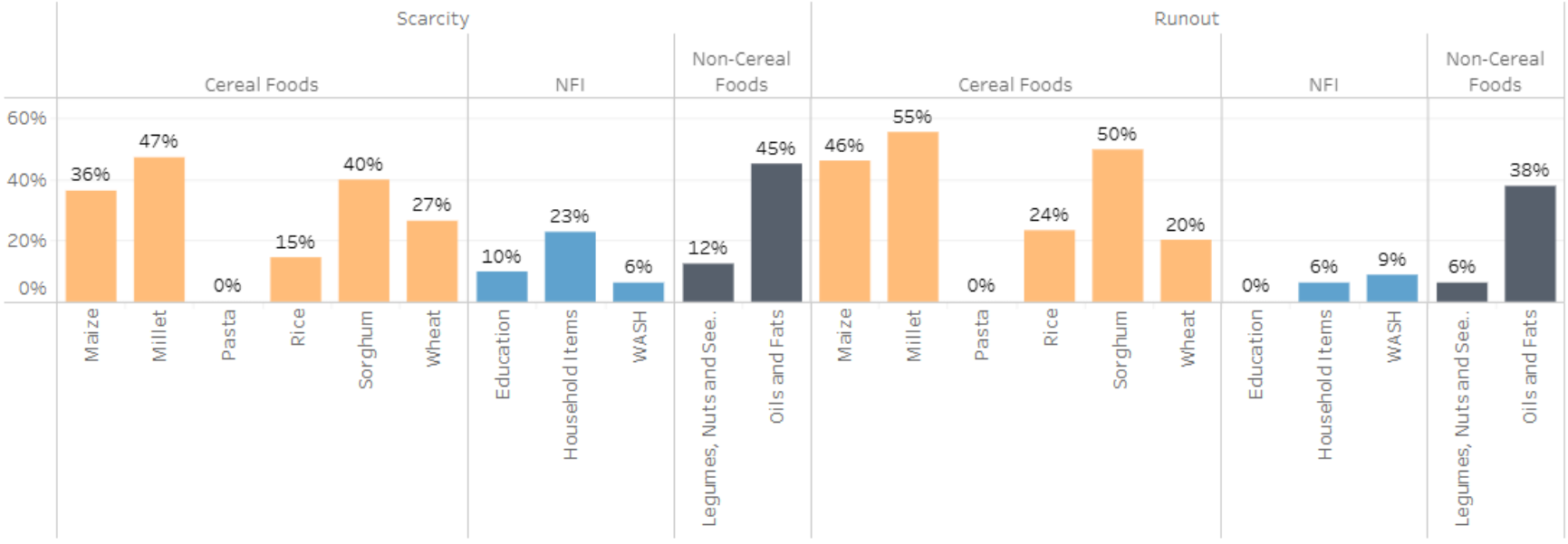
Availability
average % mentioned:



Availability - regionally
average % mentioned:

Commodities	BRONG AHAFO	NORTHERN	UPPER WEST	VOLTA
Scarce, Cereals	27%	44%	27%	82%
Scarce, NFI	16%	40%	19%	81%
Scarce, other Foods	23%	52%	41%	80%
Run out - Cereals	15%	52%	24%	82%
Run out -NFI	4%	28%	20%	65%
Run out- other Foods	18%	43%	32%	78%

Availability
average % mentioned, if yes to previous
NOTE: these are %s reported based on the traders who sell the commodity,



FINDINGS BY MFI DIMENSION : PRICE

Price was the dimension which had the lowest average national score. Volta region had the highest regional average score of 4.2, followed by Northern (3.7), Brong Ahafo (2.6) and Upper West (0.7).

Overall,

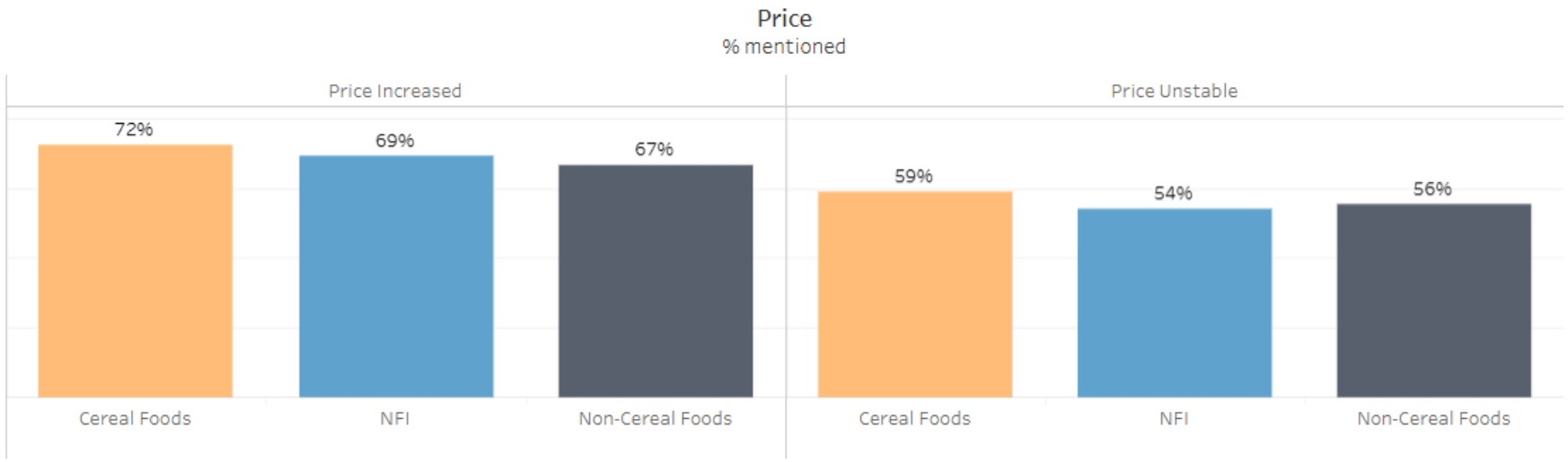
- 72% of assessed markets have experienced price increase in cereals
- 67% of assessed markets have experienced price increase in other foods
- 69% of assessed markets have experienced price increase in NFIs

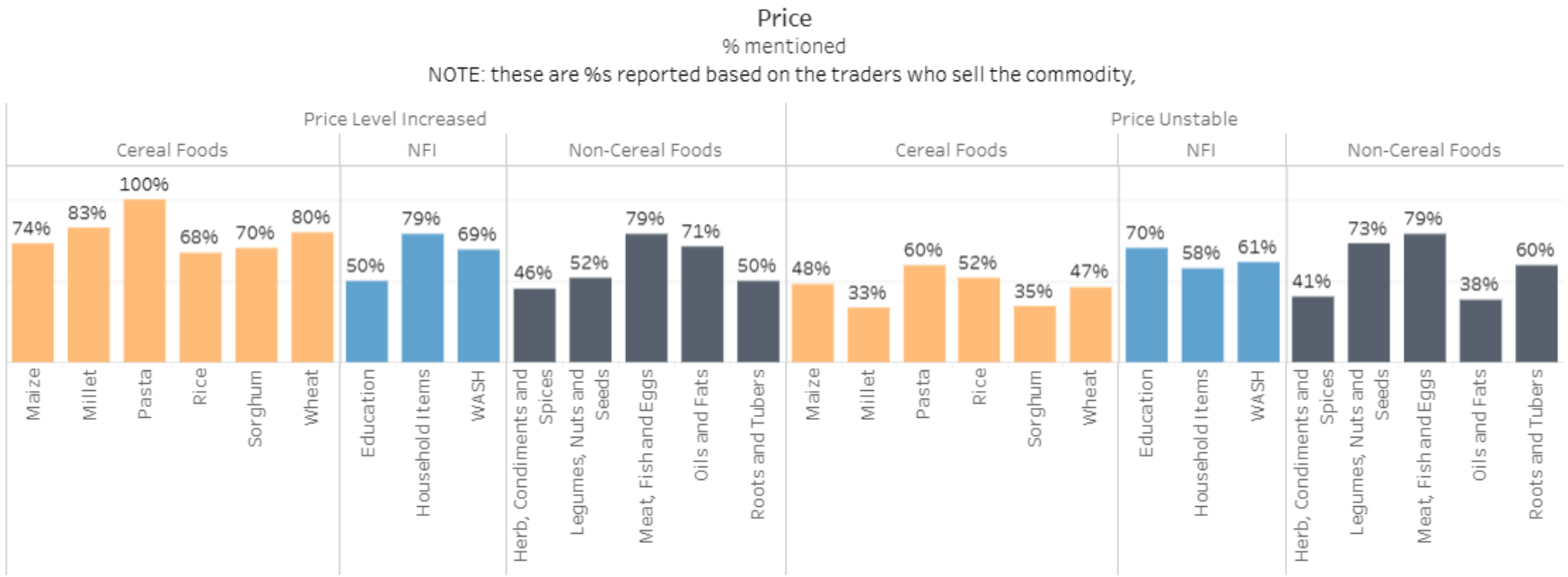
- 59% of assessed markets reported that cereal prices are unstable
- 56% of assessed markets reported that other food prices are unstable
- 54% of assessed markets reported that NFI prices are unstable

What is concerning that 10 out of the 32 markets assessed scored a 0 on the price dimension, indicating major issues of price affordability, stability and predictability across all commodity types.

For all markets with low scores on this dimension, close price/market monitoring will be key if CBT operations are to be considered. Other possible risk mitigation measures include frequent review of MEB value and increasing the frequency of distribution to adjust the value based on market price fluctuations.

RESULTS FOR ALL MARKETS





FINDINGS BY MFI DIMENSION : SUPPLY CHAIN RESILIENCE

Supply chains of assessed markets are fairly resilient, with the national average score being 8.8.

With the exception of Sang market in Northern, all markets scored 10 on supply chain responsiveness, indicating that they are able to respond to shocks in demand.

In terms of vulnerability to disruptions, with the exception of Bimbilla and Daboya markets in Northern, all markets scored well in terms of node complexity indicating that traders have more than 1 supplier or their different commodites. Most markets also scored well in terms of node criticality, indicating that traders in general do not rely on a single supplier for their commodities. Issues with node density were what brought the scores down for several markets, with traders reporting that most of their suppliers are geographically located in the same place, rendering them more vulnerable to (natural) disaster-related disruptions.

Resilience of Supply Chains

Node Density: refers to the geographical spacing of nodes within a supply chain network; Node Complexity: refers to having more than one supplier; Node criticality: refers to the presence of a single supplier upon which most traders rely upon. Explore in the table below what the median trader reports.

		Node Complexity			Node Criticality			Node Density			Respon..	Respon..
Adm1Name	MarketName	Cereals	Other Food	NFI	Cereals	Other Food	NFI	Cereals	Other Food	NFI	Current stock	Lead time
BRONG AHAFO	Bongase	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Busunya	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
	Drobo	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Jema	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No	Yes	Yes
	Nsawkaw	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Prang	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes
	Sampa	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
NORTHERN	Bima	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	Yes	Yes
	Bimbilla	Yes	Yes	No	Yes	Yes	No	No	No	Yes	Yes	Yes
	Bole	Yes	Yes	Yes	No	Yes	Yes	No	Yes	No	Yes	Yes
	Buipe	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
	Daboya	Yes	Yes	No	Yes	Yes	Yes	Yes	No	No	Yes	Yes
	Damongo	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Katinga	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
	Kumbungu	Yes	Yes	Yes	Yes	No	Yes	No	No	No	Yes	Yes
	Lungni	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	Yes	Yes
	Salaga	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	Yes	Yes
	Sang	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
	Savelugu	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Sawla	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Tamale Aboabo	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	Yes	Yes
	Tatale	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Yendi	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Zabzugu	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
UPPER WEST	Funsi	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	Yes	Yes
	Jirapa	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Moyiri	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	Yes	Yes
	Wa	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Wichau	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
VOLTA	Dambai	Yes	Yes	Yes	Yes	No	Yes	Yes	No	No	Yes	Yes
	Kpassa	Yes	Yes	Yes	Yes	No	No	No	No	No	Yes	Yes
	Nkwanta	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes

Note:

The question verbatim for **Node Complexity** reads: *"D2.2 Do you currently have more than 1 supplier for: [Cereal food, other food, and non-food items]?"*

The question verbatim for **Node Criticality** reads: *"D2.3 Does your business mostly rely upon a single supplier for: [Cereal food, other food, and non-food items]?"*

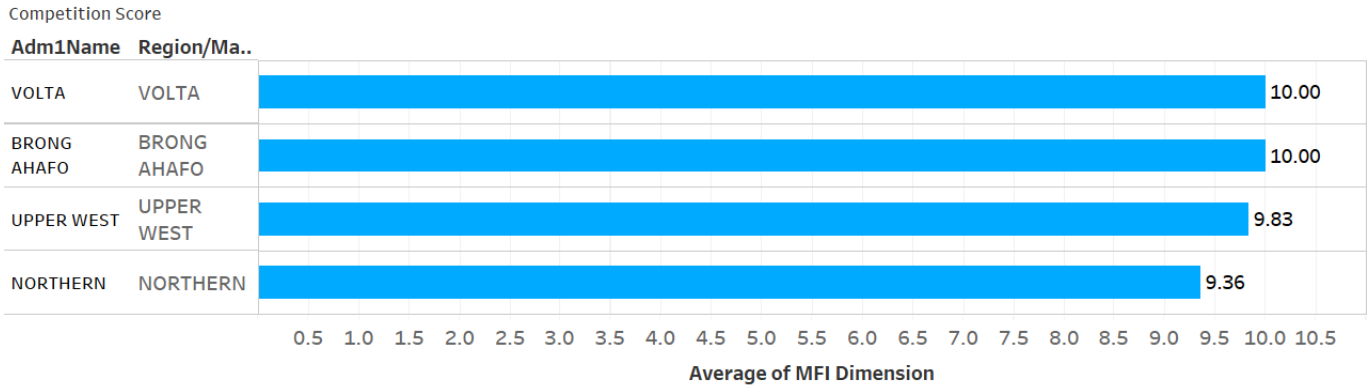
The question verbatim for **Node Density** reads: *"D2.1 Are most of your suppliers geographically located in the same place for: [Cereal food, other food, and non-food items]?"*

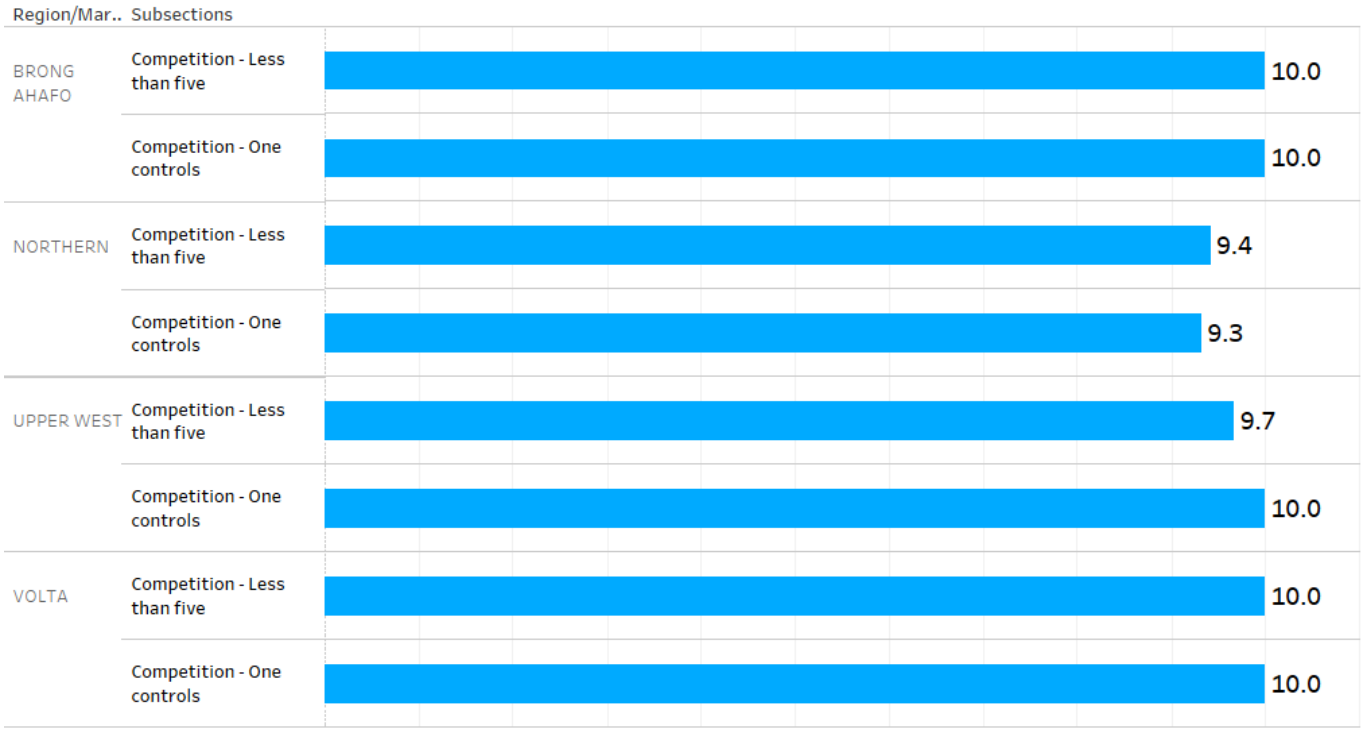
In the table above, "Yes" refers to a median of 1 indicating resilience and "No" refers to a median of 0 indicating lower resilience.

FINDINGS BY MFI DIMENSION : COMPETITION

Competition was the dimension which had the 2nd highest national average score of 9.6. With the exception of 3 markets (Bimbilla and Yendi in Northern and Moyiri in Upper West), all markets scored a 10 on this dimension. This indicates that these markets are not too concentrated or reliant on a single supplier (i.e. low risk of threat to market functionality).

Yendi market scored a 0 on this dimension, indicating issues of having less than 5 traders for each of the commodity groups as well as the presence of a monopoly for each of the commodity groups. If CBT is to be considered in this market, possible mitigation measures include vouchers with price controls and trying to introduce new traders to the area if possible.





FINDINGS BY MFI DIMENSION : INFRASTRUCTURE

As is common in markets in West Africa, the assessed markets scored relatively low on the infrastructure dimension, with a country overall score of 5.7. The Volta region had the highest regional average score of 7.4, followed by Upper West region (6.3), Brong Ahafo (5.6) and Northern (5.3).

In terms of shop conditions, the majority of markets reported "medium" category with some structure damages and minor maintenance issues as well as some shops in "good" or "poor" categories.

In terms of market infrastructure features, nearly all assessed markets have reliable communication network and walkways and emergency exits. The top 3 limitations were lack of 1) a closed sewage system (which was only available in 9 assessed markets), 2) toilets (only available in 15 assessed markets), 3) water availability (manual pump or tap in close proximity of each vendor, i.e. max 5 min walking) only available in 15 assessed markets.

		Conditions			Features						
Adm1Name	MarketName	Good	Medium	Poor	Closed sewage..	Network	Shelter	Toilet	Walka..	Waste Collectio..	Water
BRONG AHAFO	Bongase	No	Yes	Yes	No	No	Yes	No	Yes	No	Yes
	Busunya	No	Yes	No	No	No	Yes	No	Yes	No	Yes
	Drobo	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Jema	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	Yes
	Nsawkaw	Yes	Yes	Yes	No	No	Yes	No	Yes	Yes	No
	Prang	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	No
	Sampa	Yes	Yes	No	No	Yes	Yes	No	Yes	Yes	No
NORTHERN	Bima	Yes	No	No	No	Yes	Yes	No	Yes	Yes	Yes
	Bimbilla	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
	Bole	Yes	No	No	No	Yes	Yes	No	Yes	Yes	No
	Buipe	Yes	Yes	Yes	No	Yes	No	Yes	No	No	No
	Daboya	No	No	Yes	No	Yes	No	No	No	No	Yes
	Damongo	Yes	Yes	No	Yes	Yes	No	Yes	Yes	Yes	Yes
	Katinga	No	Yes	No	No	Yes	No	No	No	No	No
	Kumbungu	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes	No
	Lungni	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
	Salaga	Yes	Yes	Yes	No	Yes	Yes	No	Yes	No	Yes
	Sang	No	Yes	Yes	No	Yes	No	No	Yes	No	No
	Savelugu	No	No	Yes	No	Yes	Yes	Yes	Yes	No	No
	Sawla	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	No
	Tamale Aboabo	Yes	No	No	Yes	Yes	Yes	Yes	Yes	Yes	No
	Tatale	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No
	Yendi	No	Yes	No	No	Yes	No	No	Yes	No	No
	Zabzugu	No	No	Yes	No	Yes	No	No	Yes	No	No
UPPER WEST	Funsi	No	Yes	No	No	Yes	Yes	No	Yes	No	Yes
	Jirapa	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No	Yes
	Moyiri	No	Yes	No	No	Yes	Yes	Yes	Yes	Yes	No
	Wa	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
	Wichau	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes
VOLTA	Dambai	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes	Yes
	Kpassa	No	Yes	No	No	Yes	Yes	Yes	Yes	Yes	No
	Nkwanta	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note:

Conditions: The combination of "Yes", "No" attempts to give a description of the state of traders. As an example, the combination "Yes" ,"No", "No" (Order matters!) can be interpreted as the best one as it depicts a market where all stores are in an excellent condition.

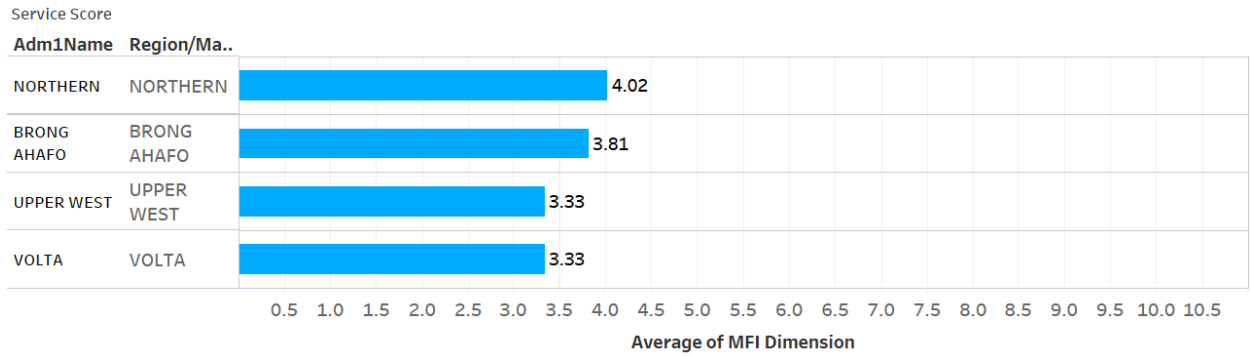
Features: The table displays a "Yes" if the considered features are present in a market, a "No" otherwise.

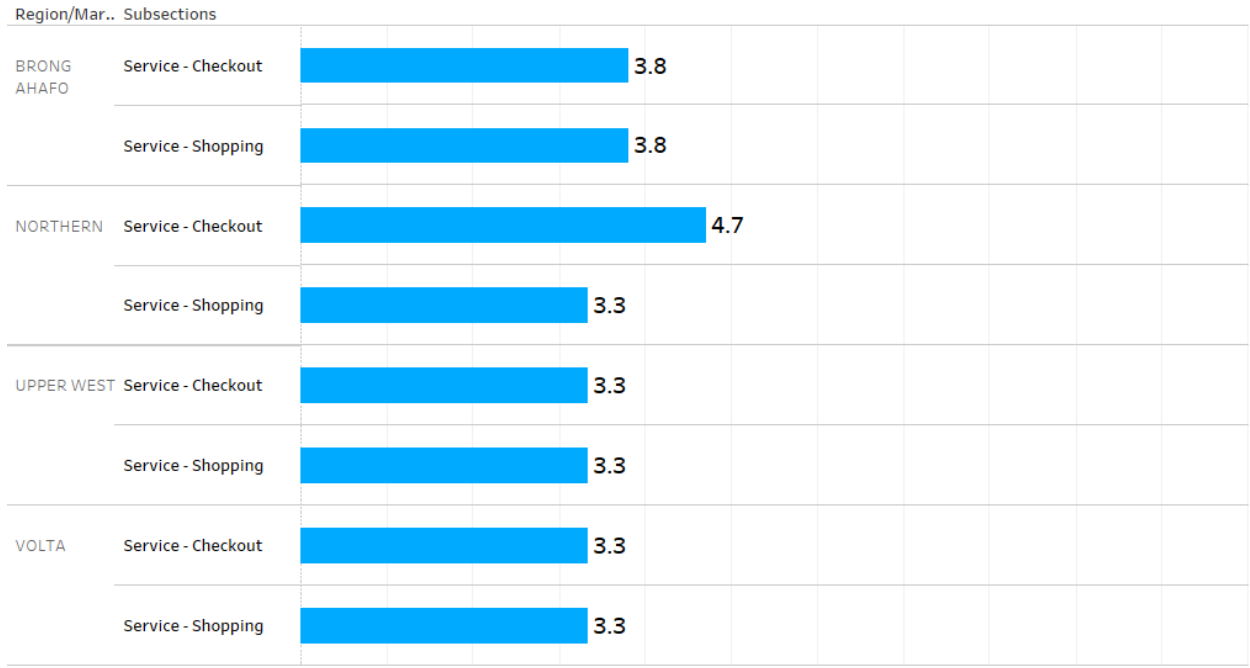
FINDINGS BY MFI DIMENSION : SERVICE

As is common in markets in West Africa, the assessed markets scored relatively low on the service dimension.

- All assessed markets reported that in general, 1) itemized receipts are not automatically issued and 2) it is not possible to purchase from the shop remotely (e.g. online shopping).
- 31 out of 32 markets assessed reported that price tags are not visibly displayed (printed or handwritten)
- 60% of markets assessed (19 out of 32) reported that on average, the waiting time at checkout is normally longer than 10 minutes.
- 16% of markets assessed (5 out of 32) reported on average, shops do not accept more than one form of payment.

Bongase market in Brong Ahofo region and Jirapa and Wichau markets in Upper West had the lowest score of 1.7.





FINDINGS BY MFI DIMENSION : FOOD QUALITY

The overall country score for the food quality dimension is 8.6, with Brong Ahofo region having the highest average score of 10 and Volta region having the lowest average score of 8.1

In general, as shown on the table, the top 3 issues were

1) refrigeration was not always working (28% of markets)
2) meats/poultry were not always refrigerated (22% of markets)
3) food not separated/well protected from contaminants (16% of markets)

This exposes consumers to a health risk. Markets such as Buipe, Daboya, Salaga, Sang in Northern region which only have 50-65% of the essential food quality features may need mitigation measures such as mobilizing local authorities to build retailer capacity and improve compliance with local food safety regulations, contract traders operating in premises with adequate food safety standards and offering food safety/quality certification to targeted retailers.

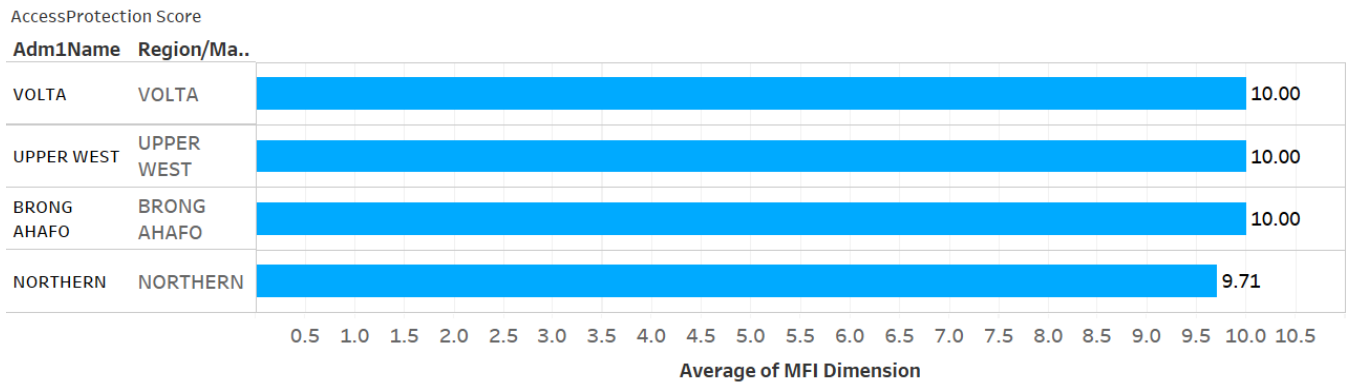
Note: The table above displays a "Yes" if the analysed feature of Food quality is predominantly present in a market, a "No" otherwise. Considering the results, find the % Essential Food quality Features to summarize the information.

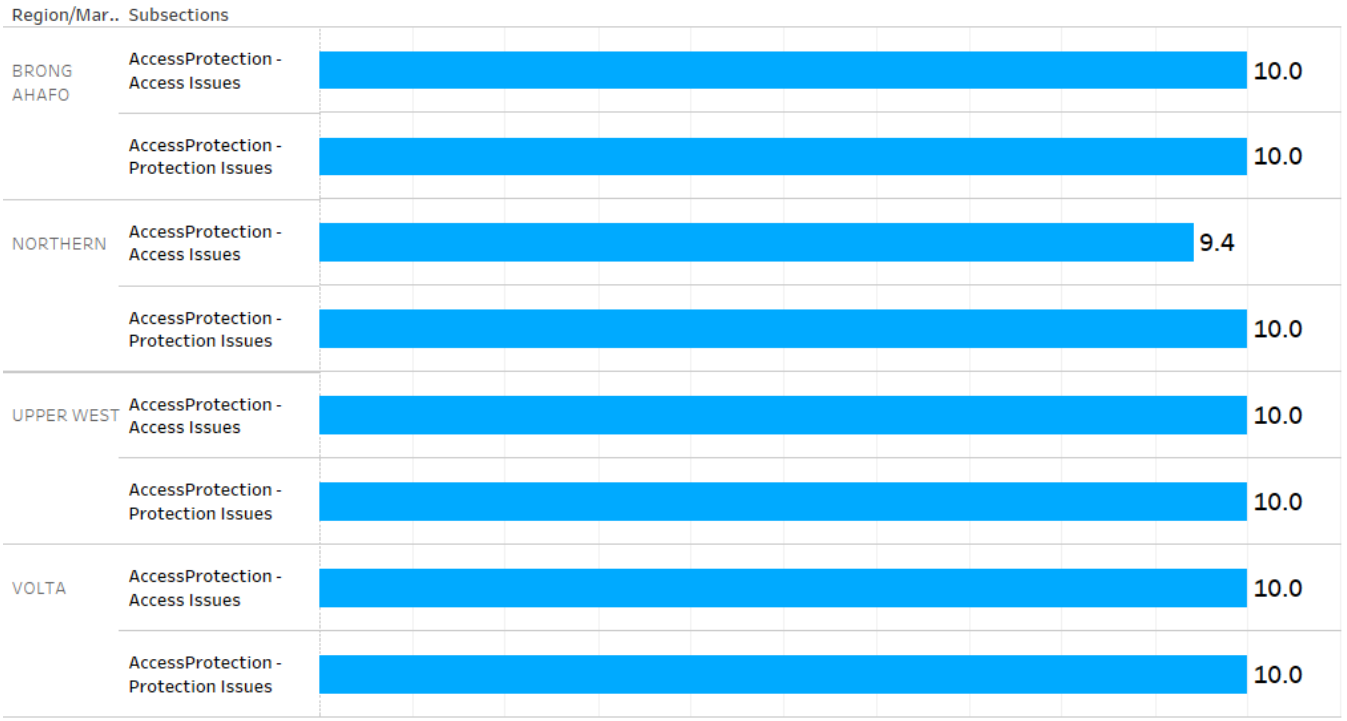
Adm1Name	MarketName	All goods are labelled an..	Food has no visible sign ..	Food separated f..	Fresh separated f..	Meats/ Poultry Foo..	Plastic packaged f..	Prepackaged intact&labe..	Refrigeraion always wor..	% Essential Food Quality Features
BRONG AHAFO	Bongase	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
	Busunya	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
	Drobo	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
	Jema	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
	Nsawkaw	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
	Prang	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
	Sampa	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
NORTHERN	Bima	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
	Bimbilla	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	88%
	Bole	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
	Buipe	Yes	Yes	No	Yes	No	No	No	Yes	50%
	Daboya	No	Yes	No	Yes	No	Yes	Yes	No	50%
	Damongo	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
	Katinga	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	88%
	Kumbungu	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	88%
	Lungni	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
	Salaga	Yes	Yes	No	Yes	No	Yes	Yes	No	63%
	Sang	Yes	Yes	No	Yes	No	Yes	Yes	No	63%
	Savelugu	Yes	Yes	No	Yes	Yes	Yes	Yes	No	75%
	Sawla	Yes	Yes	Yes	Yes	No	Yes	Yes	No	75%
	Tamale Aboabo	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
	Tatale	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
	Yendi	Yes	Yes	Yes	Yes	No	Yes	Yes	No	75%
	Zabzugu	Yes	Yes	Yes	Yes	No	Yes	Yes	No	75%
UPPER WEST	Funsi	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	100%
	Jirapa	No	No	Yes	Yes	Yes	Yes	Yes	Yes	75%
	Moyiri	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	88%
	Wa	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	88%
	Wichau	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	88%
VOLTA	Dambai	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	88%
	Kpassa	No	Yes	Yes	Yes	Yes			No	50%
	Nkwanta	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	88%

FINDINGS BY MFI DIMENSION : ACCESS & PROTECTION

This dimension had the highest overall country score of 9.8. All markets scored a full 10 for both access and protection, with the exception of Kumbungu market in Northern. This indicates that aside from Kumbungu, all markets are accessible to all consumers, with no protection issues.

Kumbungu market reported seasonal access issues, as well as difficulty of access due to being far from major road networks; it did not report any protection issues.





FINDINGS BY MFI DIMENSION : CONCLUSION

CONCLUSION

The analysis of market functionality through the different dimensions of the MFI shows that most assessed markets were well supplied with a good variety of food and non-food items, and with the exception of one (1) market, there was little concern for food quality. Overall, supply chains were also deemed resilient with sufficient competition within markets. All markets also scored relatively well in terms of access and protection.

Furthermore, it was found that the three (3) main weakest market dimensions across nearly all assessed markets were Price, Infrastructure and Service. The low scores obtained by almost all markets on these dimensions brought down the overall MFI scores. In all markets, there was reportedly an issue with higher and volatile price trends. This can be attributed to high inflation and depreciating of the Cedi vis-à-vis the USD this year, further compounded by the recent dry spell in July and August 2024, which has resulted in significant production deficit. Rate of food inflation in the month of November 2024 was 25.9%[1] which remains at all time high compared to historic trends over the last two decades. As at January 2022, the month average inter-bank Bank exchange was (GHC/US$) 6.01 as compared to November 2024 month average of (GHC/US$) 15.98[2]. Comparatively, the exchange rate loss between both months amount to 162%.

Infrastructure issues point towards possible concerns of poor food storage conditions, however, it must be noted that in general, markets in West Africa tend to score low on the Infrastructure and Service dimensions (i.e. low market maturity).

The results can be visualized in more details through the following links: Workbook: MFI Toolbox and Workbook: MFI Dashboard

1 Ghana Statistical Service (November, 2024).
2 Bank of Ghana (November, 2024).

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